WEEK 1: Networking Fundamentals | Days: 1-7

Day 4 Tasks—Subnetting Basic

- **Solve 3 subnetting problems manually (show all working).**

Subnetting Procedure (Step-by-Step)

Follow this manual procedure to solve any sub netting problem:

The 5-Step Method

Step 1: Identify what you have

- IP address (e.g., 192.168.1.0)
- Subnet mask or CIDR (e.g., /24 or 2a55.255.255.0)

How many subnets or hosts needed?

Step 2: Find the mask in binary.

- Write the subnet mask in binary (1s = network, 0s = host)

Step 3: Calculate bits borrowed**

- For more subnets = borrow host bits (turn 0s into 1s)
- Formula: 2^n = number of subnets (n = bits borrowed)

Step 4: Calculate hosts per subnet

- Formula: 2^h - 2 = usable hosts (h = remaining host bits)

- Subtract 2 for network address and broadcast address

Step 5: Find subnet ranges (increment)

- Increment = 256 - last non-zero octet in mask
- List each subnet: network, first host, last host, broadcast

Problem 1: Basic Subnetting

Question: we have `192.168.1.0/24`. You need 4 subnets. Find the new mask, subnet ranges, and hosts per subnet.

Solution:

Step 1: Given IP = 192.168.1.0, Mask = /24 (255.255.255.0)

Step 2: Binary mask = `11111111.11111111.11111111.00000000`

Step 3: Need 4 subnets → 2^n = 4 → n = 2 bits to borrow

- **New mask:** Borrow 2 host bits
- **Binary:** 11111111.11111111.11111111.11000000
- **New CIDR**: /26

New subnet mask: 255.255.255.192

Step 4: Host bits left = 8 - 2 = 6 bits

Usable hosts = 2^6 - 2 = 64 - 2 = 62 hosts per subnet

Step 5: Increment = 256 - 192 = 64

Subnet ranges:

Subnet	Network	First Host	Last Host	Broadcast
1	192.168.1.0	192.168.1.1	192.168.1.62	192.168.1.63
2	192.168.1.64	192.168.1.65	192.168.1.126	192.168.1.127
3	192.168.1.128	192.168.1.129	192.168.1.190	192.168.1.191
4	192.168.1.192	192.168.1.193	192.168.1.254	192.168.1.255

Answer: New mask = /26 (255.255.255.192), 4 subnets, 62 hosts each

Problem 2: Subnet Based on Host Requirement

Question: we have `172.16.0.0/16`. You need subnets with at least 500 hosts each. Find the new mask and how many subnets you get.

Solution:

Step 1: Given IP = 172.16.0.0, Mask = /16 (255.255.0.0)

Step 2: Binary mask = `11111111.11111111.00000000.00000000`

- Host bits total = 16

Step 3: Need 500 hosts per subnet

- Formula: 2^h - 2 ≥ 500

- 2^9 = 512 - 2 = 510 hosts ✅ (500 fits)

- 2^8 = 256 - 2 = 254 hosts ❌ (too small)

- So h = 9 host bits needed

Step 4: Borrowed bits = Total host bits - Host bits needed

- Borrowed = 16 - 9 = 7 bits

New mask = /16 + 7 = /23

New subnet mask = 255.255.254.0

Step 5: Number of subnets = 2^7 = 128 subnets

Step 6: Increment = 256 - 254 = 2 (in the 3rd octet)

- Example subnets: 172.16.0.0/23, 172.16.2.0/23, 172.16.4.0/23...

Answer: New mask = /23 (255.255.254.0), 128 subnets, 510 hosts each

Problem 3: Find Subnet of a Specific IP

Question: IP address `10.50.100.75` with subnet mask `255.255.240.0`. What subnet is this IP in? What is the network address and broadcast address?

Solution:

Step 1: IP = 10.50.100.75, Mask = 255.255.240.0

Step 2: Convert mask to binary:

- 255.255.240.0 = 11111111.11111111.11110000.00000000

Step 3: Find which octet changes (the "interesting octet")

- First 2 octets are 255 → stay same as IP
- 3rd octet = 240 (interesting octet)
- 4th octet = 0 → all zeros in host portion

Step 4: Increment = 256 - 240 = 16 (in the 3rd octet)

Step 5: Find which subnet range contains 100 (3rd octet of IP)

- Subnets in 3rd octet: 0, 16, 32, 48, 64, 80, 96, 112, 128...
- 100 falls between 96 and 112
- Network 3rd octet = 96

Step 6: Build the network address

- Network = 10.50.96.0 (3rd octet = 96, 4th octet = 0)

Step 7: Find broadcast address

- Next network's 3rd octet = 112
- Broadcast = 10.50.111.255 (112 - 1 = 111 for 3rd octet, all 1s for host)

Step 8: Host range

- First host = 10.50.96.1
- Last host = 10.50.111.254

Answer:

- Network address: 10.50.96.0
- Broadcast address: 10.50.111.255
- This IP is in subnet: 10.50.96.0/20

Quick Reference Table

CIDR	Subnet Mask	Increment	Hosts
/24	255.255.255.0	1	254
/25	255.255.255.128	128	126
/26	255.255.255.192	64	62
/27	255.255.255.224	32	30
/28	255.255.255.240	16	14
/29	255.255.255.248	8	6
/30	255.255.255.252	4	2

How to Perform This Practically (Hands-on)

What you need:

- Paper and pen (or notepad)
- Calculator (optional)

Step-by-step practical exercise:

1. Write down the problem** exactly as given
2. Draw a table with 4 columns: Subnet # | Network | First Host | Last Host | Broadcast.
3. Convert to binary (practice writing 8 bits for each octet)
4. Calculate increment (256 - interesting octet value)
5. List subnets by adding increment repeatedly
6. Verify your work:
   - Network address = all host bits = 0
   - Broadcast = all host bits = 1
   - Hosts between them

➢ Task: Calculate Subnet Masks for Given Requirements?

The Basic Formula

Two types of requirements:

1. Number of subnets needed→ Borrow bits from host portion.
2. Number of hosts per subnet needed → keep enough host bits.

Procedure (Simple 4 Steps)

Step 1: Start with the default mask based on IP class

- **Class A**: /8 (255.0.0.0)
- **Class B:** /16 (255.255.0.0)
- **Class C:** /24 (255.255.255.0)

Step 2: Determine how many bits to borrow or keep

- **For subnets: 2^n ≥ number of subnets needed**
- **For hosts: 2^h - 2 ≥ number of hosts needed**

Step 3: Calculate the new mask

- New CIDR = Default CIDR + n (bits borrowed)

- Or New CIDR = 32 - h (host bits kept)

Step 4: Convert CIDR to dotted decimal

- Write binary: n ones, then zeros
- Convert each 8-bit group to decimal

Example 1: Need 8 Subnets from Class C

Requirement: 8 subnets from 192.168.1.0

Solution:

- Default mask = /24
- 2^n ≥ 8 → n = 3 bits borrowed
- New CIDR = 24 + 3 = /27
- Binary mask: 11111111.11111111.11111111.11100000
- Dotted decimal: 255.255.255.224

Answer: 255.255.255.224

Quick Reference Table

Requirement	Default	Calculation	New Mask
2 subnets from Class C	/24	2^1 → /25	255.255.255.128
4 subnets from Class C	/24	2^2 → /26	255.255.255.192
8 subnets from Class C	/24	2^3 → /27	255.255.255.224
16 subnets from Class C	/24	2^4 → /28	255.255.255.240
32 subnets from Class C	/24	2^5 → /29	255.255.255.248
126 hosts from Class C	/24	2^7-2 → /25	255.255.255.128
62 hosts from Class C	/24	2^6-2 → /26	255.255.255.192
30 hosts from Class C	/24	2^5-2 → /27	255.255.255.224
14 hosts from Class C	/24	2^4-2 → /28	255.255.255.240

Common Mask Values (Dotted Decimal)

CIDR	Mask	Binary (last octet)
/24	255.255.255.0	00000000
/25	255.255.255.128	10000000
/26	255.255.255.192	11000000
/27	255.255.255.224	11100000
/28	255.255.255.240	11110000
/29	255.255.255.248	11111000
/30	255.255.255.252	11111100

CIDR	Mask	Binary (2nd & 3rd octet)
/8	255.0.0.0	11111111.00000000.00000000
/9	255.128.0.0	11111111.10000000.00000000
/10	255.192.0.0	11111111.11000000.00000000
/11	255.224.0.0	11111111.11100000.00000000
/12	255.240.0.0	11111111.11110000.00000000
/13	255.248.0.0	11111111.11111000.00000000
/14	255.252.0.0	11111111.11111100.00000000
/15	255.254.0.0	11111111.11111110.00000000
/16	255.255.0.0	11111111.11111111.00000000

Quick Reference Table

CIDR	Subnet Mask	Total IPs	Usable IPs	Common Use
/8	255.0.0.0	16,777,216	16,777,214	Large organizations
/16	255.255.0.0	65,536	65,534	Medium networks
/24	255.255.255.0	256	254	Small LAN (standard)
/28	255.255.255.240	16	14	Small subnets
/30	255.255.255.252	4	2	Point-to-point links
/32	255.255.255.255	1	1	Single host (loopback)

➢ Task :CIDR notation (e.g. /24, /28)

CIDR:

/24 = 255.255.255.0

/28 = 255.255.255.240

Network bits = fixed portion

Host bits = variable portion

CIDR (Classless Inter Domain Routing) is a method of IP address allocation and routing that allows more efficient use of IP addresses. Unlike traditional class-based addressing, CIDR allocates IP addresses based on a network prefix rather than a fixed class (A, B, or C).

CIDR Notation: a.b.c.d/n

n = number of bits in the network prefix.

Example: 192.168.1.0/24 first 24 bits are the network, and the remaining 8 bits are the host ID.

Why CIDR?

Classful addressing wastes IP addresses:

Class	IPs Available	Hosts	Example Wastage
A	2^{24}	$2^{24} - 2$	Too large for small orgs
B	2^{16}	$2^{16} - 2$	Wastes 49,150 hosts for 214 needed
C	2^{8}	$2^{8} - 2$	Small networks only

Problem: Organizations often need a number of hosts that do not match class sizes, leading to wastage.
Solution: CIDR allows flexible block allocation matching exact requirements.

Rules for Forming CIDR Blocks

- All IPs must be contiguous.
- Block size must be a power of 2 (2^n) simplifies network division.
- First IP of block divisible by block size least significant bits of host ID should be 0. 1/6 5

Example: If the Block size is 2 then, Host Id will contain 5 bits and Network will contain 32 - 5 = 27 bits. First IP address of the Block must be evenly divisible by the size of the block. In simple words, the least significant part should always start with zeroes in Host Id. Since all the least significant bits of Host Id is zero, then we can use it as Block Id part.

Example: Check whether 100.1.2.32 to 100.1.2.47 is a valid IP address block or not? 4

- All the IP addresses are contiguous.
- Total number of IP addresses in the Block = 16 = 2.
- 1st IP address: 100.1.2.00100000 Since, Host Id will contains last 4 bits and all the least significant 4 bits are zero. Hence, first IP address is evenly divisible by the size of the block.

All three rules are followed by this Block. Hence, it is a valid IP address block.

Advantages of CIDR

- **Efficient IP usage:** Minimizes IPv4 address wastage.
- **Flexible allocation:** Supports networks of any size.
- **Improved routing:** Aggregates addresses for faster, simpler routing.
- **Lower administrative overhead:** Easier IP and network management.

Disadvantages of CIDR

- **Complexity:** CIDR is more complex to implement and manage compared to traditional class-based addressing. Compatibility
- **Issues:** Some older network devices may not support CIDR.
- **Security Concerns:** Implementing security measures like firewall rules and access control lists can be more difficult.

CIDR was introduced primarily to

A

Increase the number of IPv4 addresses

B

Replace classful addressing and reduce IP address wastage

C

2/6 Make routing tables smaller forever

D

Support IPv6 only

In CIDR notation, 192.168.10.0/25 means

A

25 host bits, 7 network bits

B

25 network bits, 7 host bits

C

25 total bits are used

D

25 subnets are possible

What is the block size (number of IP addresses) for a /28 network?

A	**8**
B	**14**
C	**16**
D	**32**

Before CIDR, IP addresses used fixed classes (A, B, C), which wasted many addresses. CIDR allows flexible subnet size you can have a /27 (32 IPs) or /29 (8 IPs) instead of only /24 (256 IPs) or /16 (65k IPs).

➢ TASK: Difference between Network bits and Host bits.

Initially in 1980 IP address was divided into two fixed parts NID=8 bit and HID=24 bit. NID (Network ID) represents the entire network for a specified host. HID (Host ID) represents a specified user IP address within the network. So there are only 2 i.e., 256 networks are available and there are 2 i.e., 16M hosts per network. If even a small organization wants to purchase a network then it should buy one network in which there are 16M hosts that can be connected this is the drawback of this fixed representation of IP address. 8 IP addresses

are divided into 5 classes: Class A is used for big networks such as ISP networks. Class B is used by big organizations. Class C is used by small organizations. Class D no NID and HID all 28 bits are used to define the multicast address. Class E is reserved for research and future purchases.

Range of All Classes

Class A: 1.0.0.0 to 127.255.255.255

Class B: 128.0.0.0 to 191.255.255.255

Class C: 192.0.0.0 to 223.255.255.255

Class D: 224.0.0.0 to 239.255.255.255

Class E: 240.0.0.0 to 255.255.255.255

What is Network ID?

A network ID also known as NetID is a fixed bit in the IP address that represents the entire network of any host it is connected to the network. i.e., it tells the network of the host by which the host is connected. The total size of the IP address is 32 bits and the IP address is divided into four octets each of size is 8 bits. And further, divide into NID and HID. Let's discuss each class in detail.

Class A: The first octet represents NID so there are 8 bits for NID and 24 bits for HID. And the first bit of the first octet is always 0.

Class B: The first two octets represent NID so there are 16 bits for NID and 16 bits for HID. And the first two bits of the first octet are fixed i.e., 10.

Class C: The first three octets represent NID so there are 24 bits for NID and 8 bits for HID. The first three bits of the first octet are fixed i.e., 110.

Class D: All bits are fixed for NID.

Class E: All bits are fixed for NID.

The network ID uses a prefix based on its class. Classes A, B, C, D, and E use prefixes '0', '10', '110', '1110', and '1111 0', respectively. Additionally, as seen in the graphic below, each class has a different number of octets reserved for the host ID. Class A reserves 3 octets, class B reserves 2, class C reserves 1, and class D/E have no reserved octets for the host ID. Combining these two facts, we see that Classes A, B, C, D, and E use 7, 14, 21, 28, and 27 bits, respectively, for the actual network ID.

What is Host ID?

It is the fragment of an IP address that uniquely identifies a host within a network. The network ID can be found by ANDing the IP (in binary) and subnet mask (in binary). The host ID is found in the same way, except one must first invert the subnet mask (not an official term, but could that binary value the 'host mask').

Question 1: What is the network address if the destination address is 200.45.34.56 and the subnet mask is 255.255.255.240?

Solution: We know the range of class C is from 192 to 223 so the given IP address belongs to class C. As given subnet mask 255.255.255.240 To find the network address we perform ANDing operation between the

IP address and subnet mask. 200.45.34.00111000 AND 255.255.255.11110000 we get 200.45.34.48 as a network address.

Question 2: A company is granted the site address 181.56.0.0. The company needs 1000 subnets. Find the subnet mask

Solution: As we know the range of class B IP addresses in the first octet is from 128 to 191 so the given address belongs to class B. and the default subnet mask of class B is, 255.255.0.0 to get 1000 subnets we perform a log operation log 1000=10 So, we require extra 10 bits from 255.255.00000000.00000000 to get the subnet mask i.e.; 255.255.00000000.000000000 So, the required subnet mask is 255.255.11111111.11000000 or 255.255.255.192.

Conclusion

ID address is divided into two parts that is Network ID and Host ID. Network ID indicates the specific network to which the address belongs. Host ID Identifies the specific device within that network. Together, these components enable efficient routing of data packets across the networks, ensuring proper delivery to the correct device.

(The End)